
Power & Battery
File: 01_power_battery.kicad_sch
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File: 02_ec_mcu.kicad_sch
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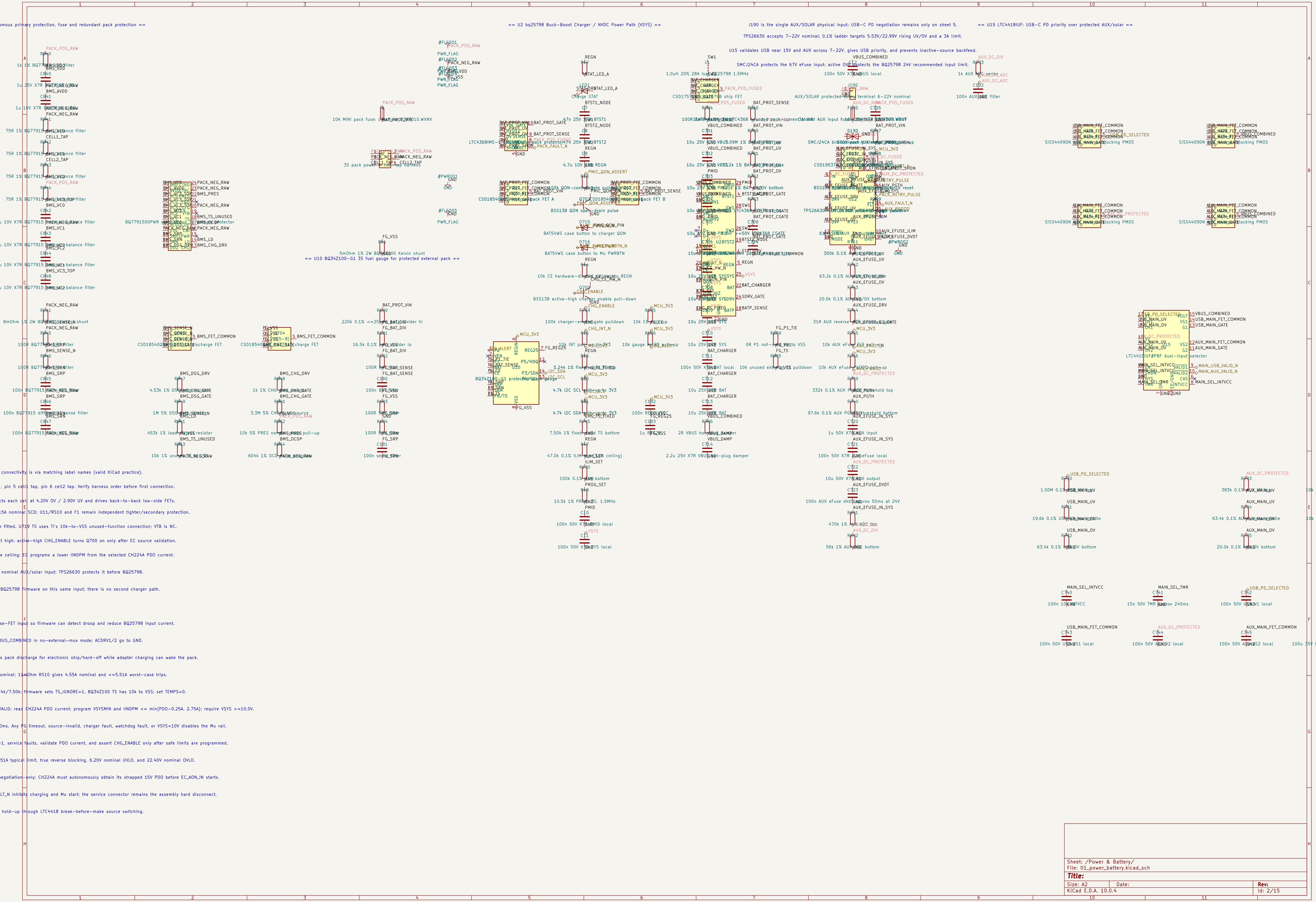
System Audio
File: 15_system_audio.kicad_sch
Gigabit Ethernet
File: 16_gigabit_ethernet.kicad_sch
Power Inputs
Sheet: /
File: ducktop2.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/15



== U2 bq25798 Buck-Boost Charger / NVDC Power Path (VSYS) ==
J190 is the single AUX/SOLAR physical input; USB-C PD negotiation remains only on sheet 5.
== U15 LTC4418IUF: USB-C PD priority over protected AUX/solar ==
TPS26630 accepts 7–22V nominal; 0.1% ladder targets 5.53V/22.99V rising UV/OV and a 3A limit.
U15 validates USB near 15V and AUX across 7–22V, gives USB priority, and prevents inactive-source backfeed.
SMCJ24CA protects the 67V eFuse input; active OVP protects the BQ25798 24V recommended input limit.
== U10 BQ34Z100-G1 3S fuel gauge for protected external pack ==
LTC4418IUF#PBF dual-input selector
connectivity is via matching label names (valid KiCad practice).
, pin 5 cell1 tap, pin 6 cell2 tap. Verify harness order before first connection.
cts each cell at 4.20V OV / 2.90V UV and drives back-to-back low-side FETs.
5A nominal SCD; U11/RS10 and F1 remain independent tighter/secondary protection.
e fitted. U719 TS uses TI's 10k-to-VSS unused-function connection; VTB is NC.
t high; active-high CHG_ENABLE turns Q700 on only after EC source validation.
e ceiling; EC programs a lower IINDPM from the selected CH224A PDO current.
nominal AUX/solar input; TPS26630 protects it before BQ25798.
BQ25798 firmware on this same input; there is no second charger path.
se-FET input so firmware can detect droop and reduce BQ25798 input current.
BUS_COMBINED in no-external-mux mode; ACDRV1/2 go to GND.
s pack discharge for electronic ship/hard-off while adapter charging can wake the pack.
ominal; 11mOhm RS10 gives 4.55A nominal and <=5.51A worst-case trips.
4k/7.50k; firmware sets TS_IGNORE=1. BQ34Z100 TS has 10k to VSS; set TEMPS=0.
VALID; read CH224A PDO current; program VSYSMIN and IINDPM <= min(PDO−0.25A, 2.75A); require VSYS >=10.0V.
0ms. Any PG timeout, source-invalid, charger fault, watchdog fault, or VSYS<10V disables the Mu rail.
1, service faults, validate PDO current, and assert CHG_ENABLE only after safe limits are programmed.
51A typical limit, true reverse blocking, 6.20V nominal UVLO, and 22.40V nominal OVLO.
egotiation-only; CH224A must autonomously obtain its strapped 15V PDO before EC_AON_IN starts.
LT_N inhibits charging and Mu start; the service connector remains the assembly hard disconnect.
hold-up through LTC4418 break-before-make source switching.
Sheet: /Power & Battery/
File: 01_power_battery.kicad_sch
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native Mu USB 3.2 Gen 2 Type−C host ports ==

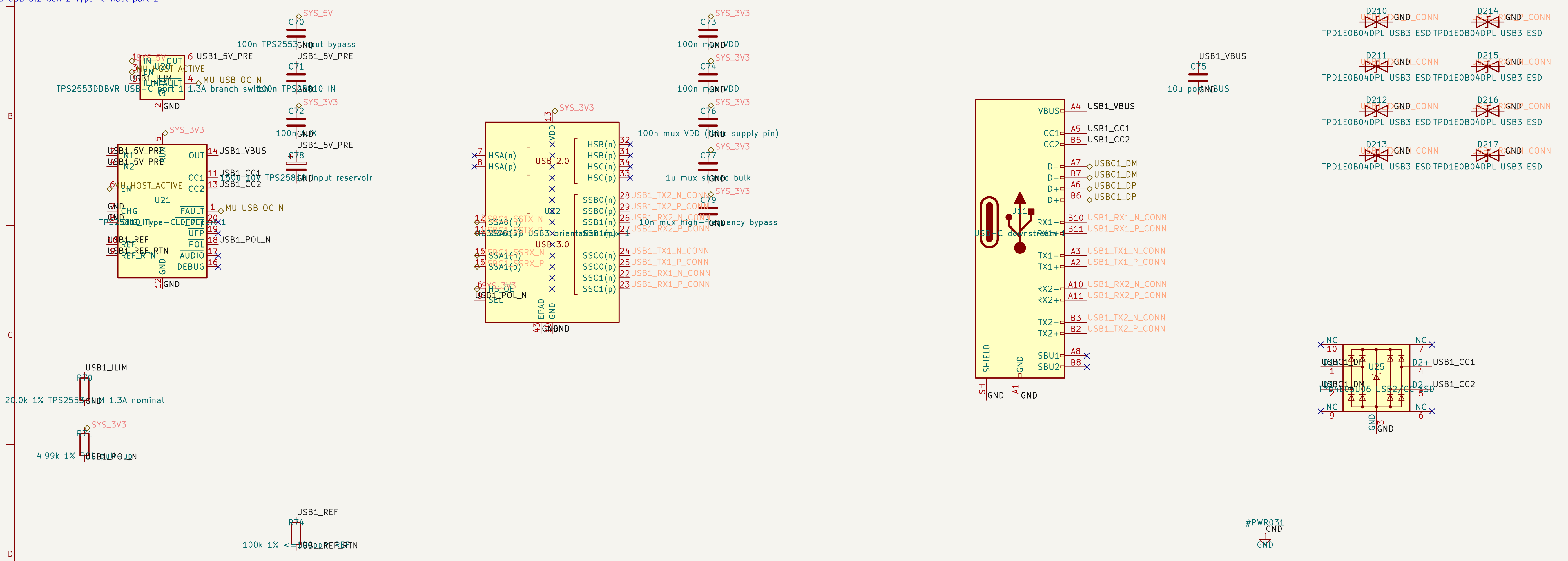
/VBUS; TI rates the HD3SS6126 high−bandwidth path up to 10Gbps.

efault USB current; branch−switch and port FAULT outputs share Mu USB_OC#.

E = PSON AND MU_12V_PG; suspend or a failed Mu rail removes VBUS.

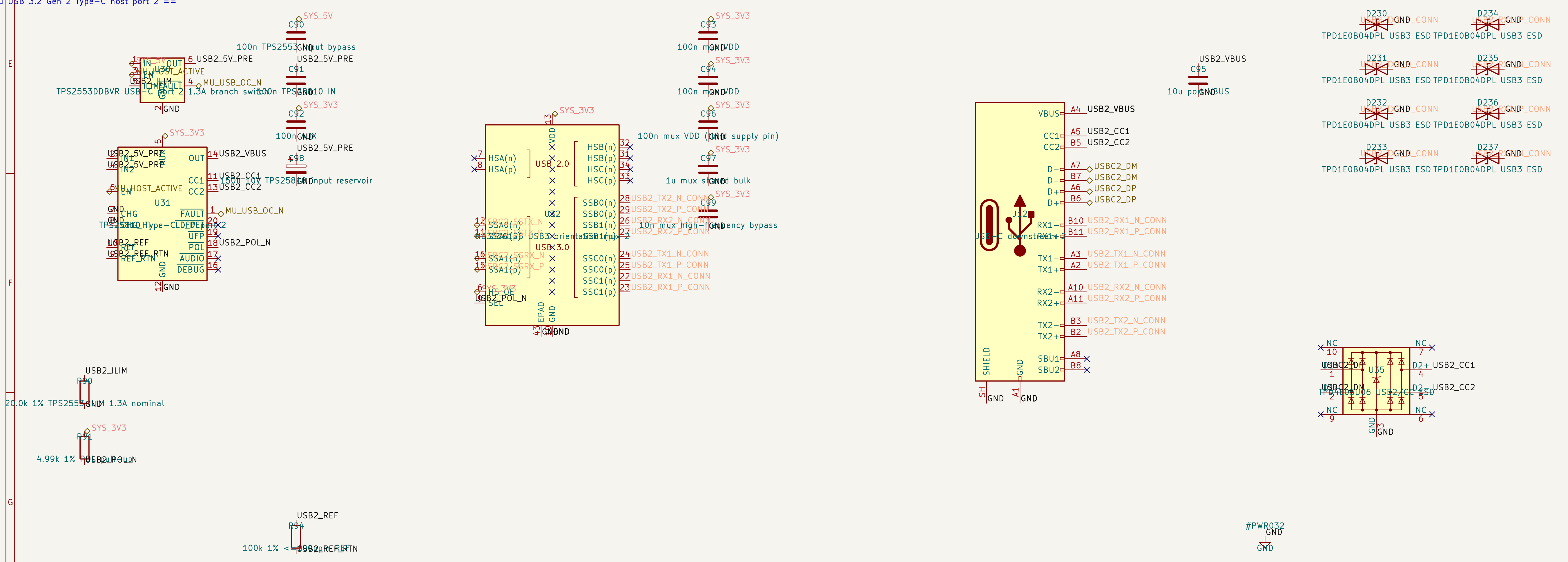
th trackpad, fan, audio, radios, and maker power; simultaneous−load validation/load shedding is mandatory.

native Mu USB 3.2 Gen 2 Type−C host port 1 ==



host, default−current advertisement; TCP0 is HDMI−A on sheet 6.

native Mu USB 3.2 Gen 2 Type−C host port 2 ==



host, default−current advertisement; TCP0 is HDMI−A on sheet 6.

Sheet: /Native USB−C I/O/
File: 04_usb_c_io.kicad_sch

Title:

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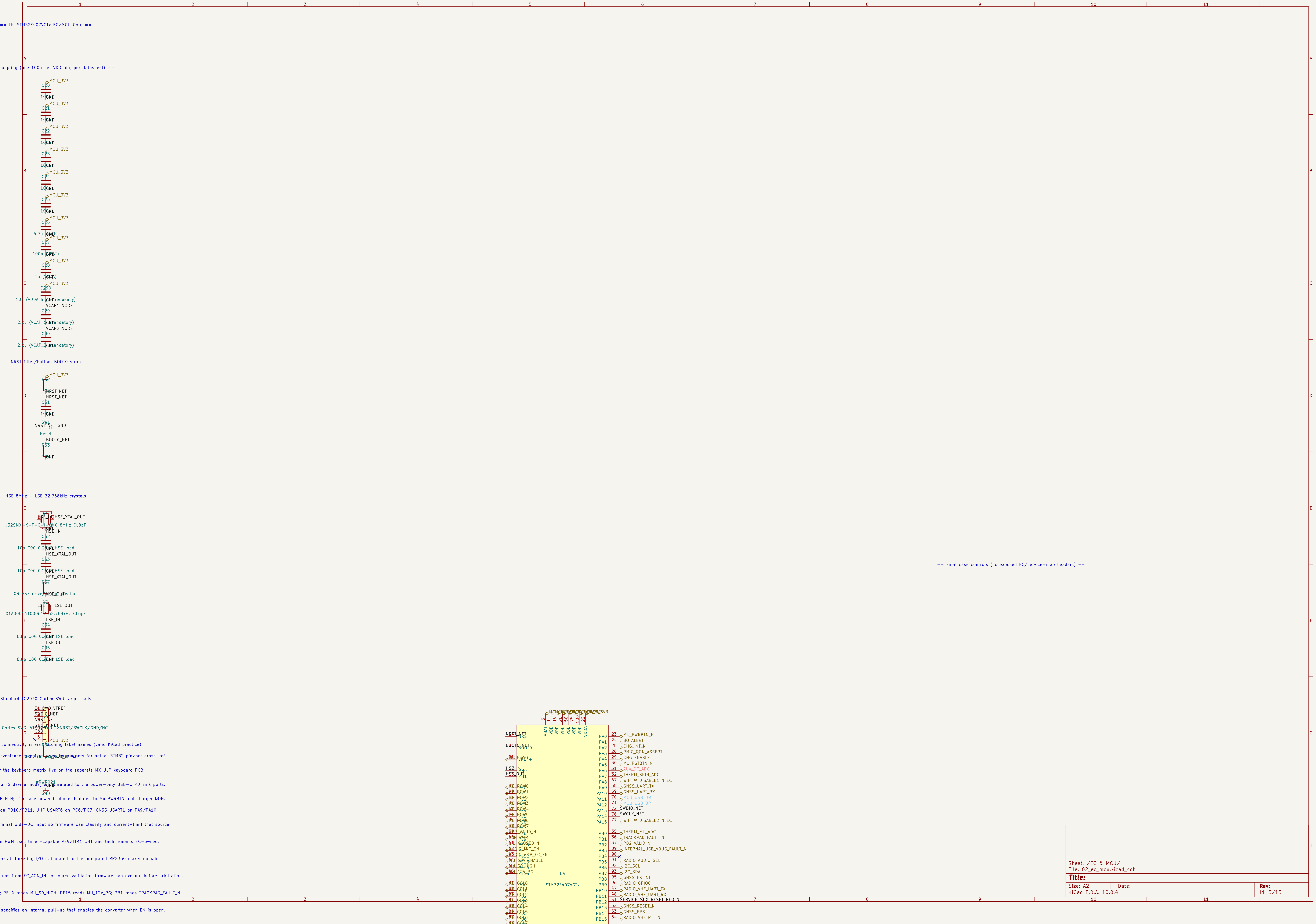


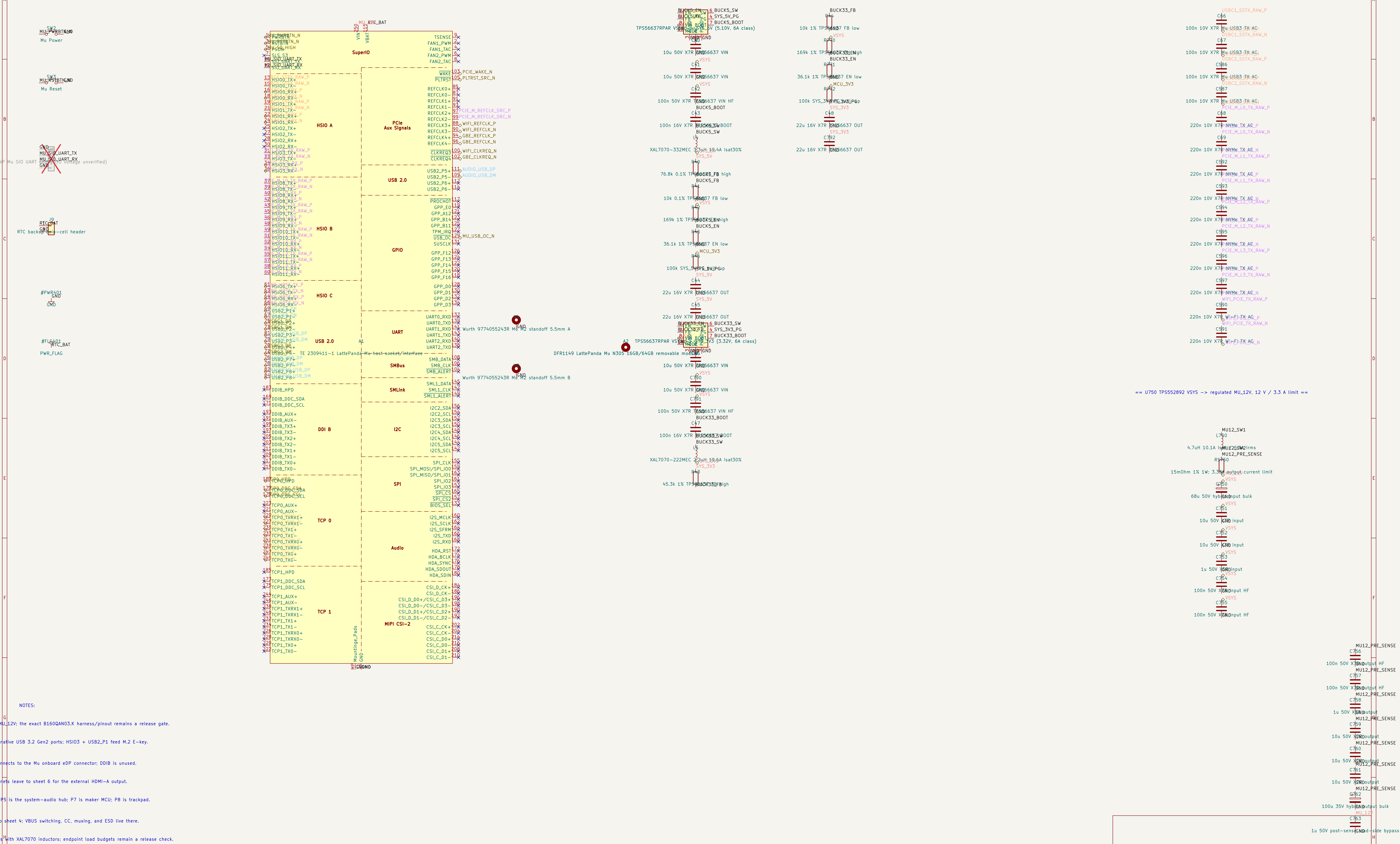
Sheet: /EC & MCU/
File: 02_ec_mcu.kicad_sch

Title:

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 5/15

== A1 LattePanda Mu carrier core ==

== Local carrier rails from VSYS ==



NOTES:

egulated MU_12V; the exact B160QAN03.K harness/pinout remains a release gate.

ependent native USB 3.2 Gen2 ports; HSIO3 + USB2_P1 feed M.2 E-key.

panel connects to the Mu onboard eDP connector; DDIB is unused.

HPD/DDC nets leave to sheet 6 for the external HDMI-A output.

SB-C 2; P5 is the system-audio hub; P7 is maker MCU; P8 is trackpad.

s leave to sheet 4; VBUS switching, CC, muxing, and ESD live there.

lass bucks with XAL7070 inductors; endpoint load budgets remain a release check.

8-11 x4 and REFCLK2; no lane reversal and no TX/RX direction swap.

rough 220n near J10; PET/PER naming is from the host perspective.

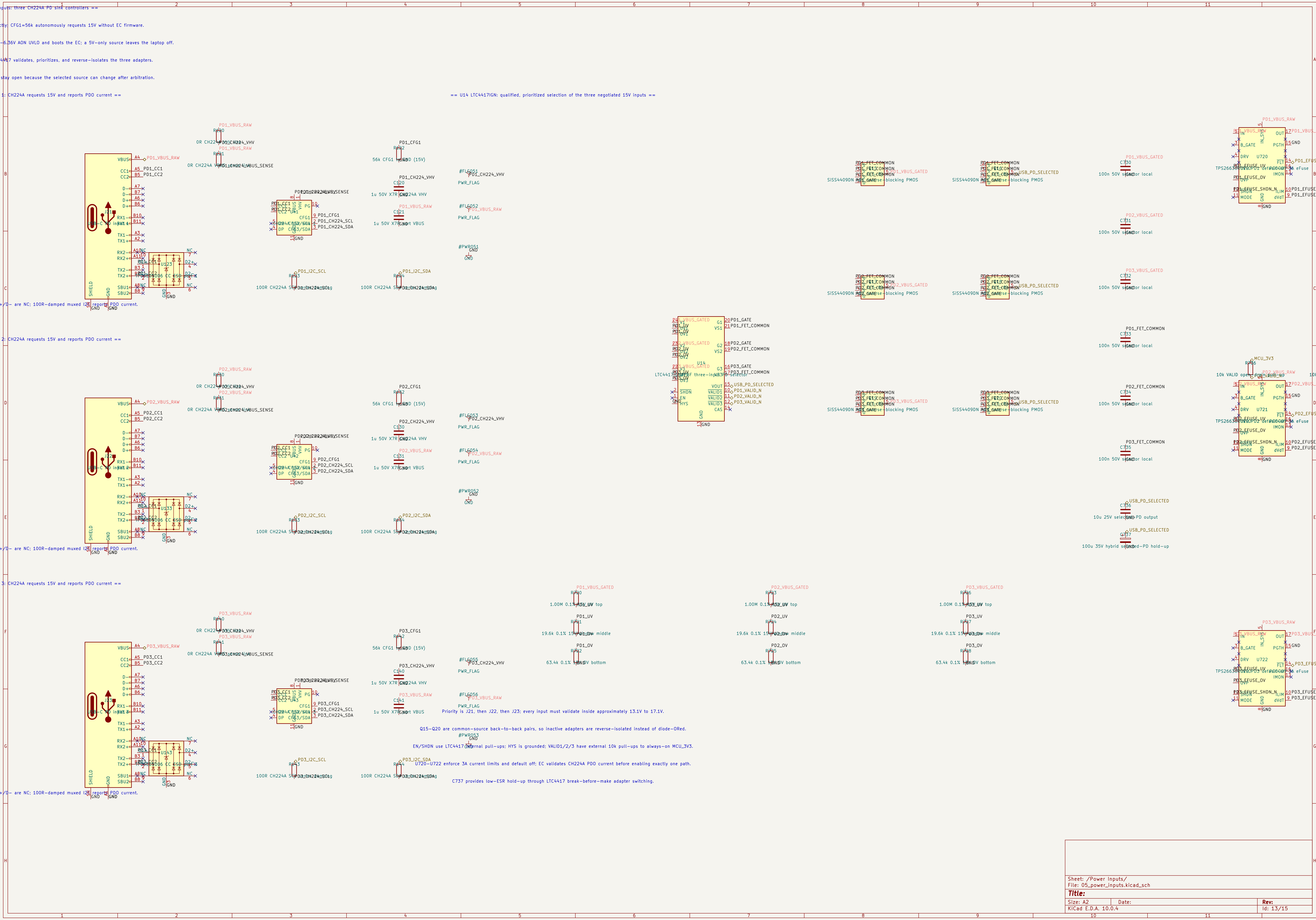
pe-C PD controller direction; trackpad consumes the former USB2_P8 spare.

WM, 3.33A nominal limit, and about 8.65-9.40V worst-case rising UVLO. Firmware requires VSYS >=10.0V.

Sheet: /Mu Carrier/
File: 03_mu_carrier.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/15



power inputs: three CH224A PD sink controllers ==
24A directly; CFG1=56k autonomously requests 15V without EC firmware.
the 6.06–6.36V AON UVLO and boots the EC; a 5V-only source leaves the laptop off.
rent; LTC4417 validates, prioritizes, and reverse-isolates the three adapters.
D+/D- stay open because the selected source can change after arbitration.
ink input 1: CH224A requests 15V and reports PDO current ==
== U14 LTC4417IGN: qualified, prioritized selection of the three negotiated 15V inputs ==
ink input 2: CH224A requests 15V and reports PDO current ==
ink input 3: CH224A requests 15V and reports PDO current ==
H224A D+/D- are NC; 100R-damped muxed I2C reports PDO current.
Priority is J21, then J22, then J23; every input must validate inside approximately 13.1V to 17.1V.
Q15–Q20 are common-source back-to-back pairs, so inactive adapters are reverse-isolated instead of diode-ORed.
EN/SHDN use LTC4417 internal pull-ups; HYS is grounded; VALID1/2/3 have external 10k pull-ups to always-on MCU_3V3.
U720–U722 enforce 3A current limits and default off; EC validates CH224A PDO current before enabling exactly one path.
C737 provides low-ESR hold-up through LTC4417 break-before-make adapter switching.
PD1_VBUS_RAW
PD2_VBUS_RAW
PD3_VBUS_RAW
PD1_VBUS_GATED
PD2_VBUS_GATED
PD3_VBUS_GATED
USB_PD_SELECTED
MCU_3V3
SISS4409DN reverse-blocking PMOS
100n 50V selector local
10u 25V selected PD output
100u 35V hybrid selected-PD hold-up
1.00M 0.1% 15V UV top
19.6k 0.1% 15V UV/OV middle
63.4k 0.1% 15V OV bottom
56k CFG1 (15V)
1u 50V X7R CH224A VHV
100R CH224A SCL
100R CH224A SDA
PWR_FLAG
GND
Sheet: /Power Inputs/
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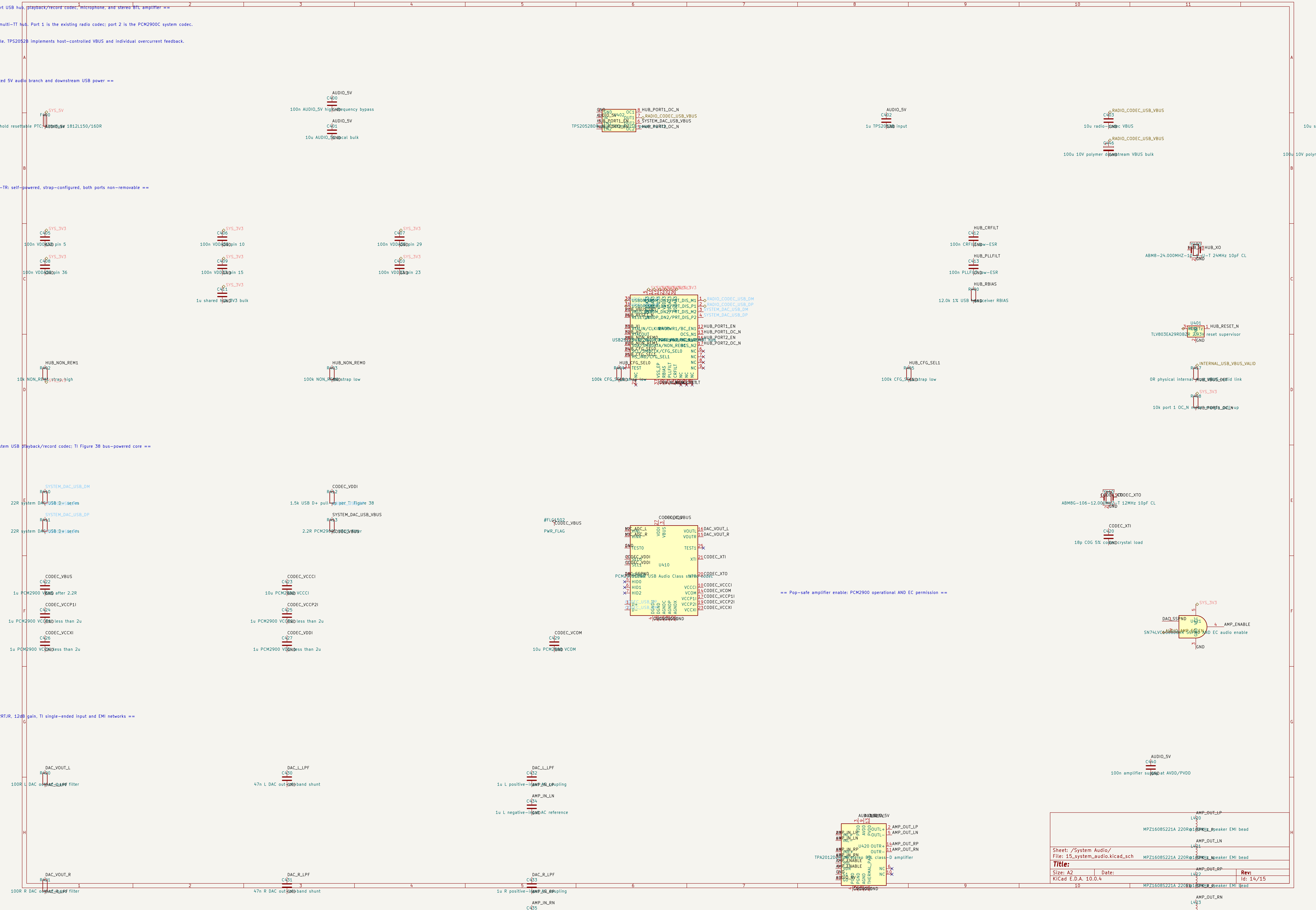
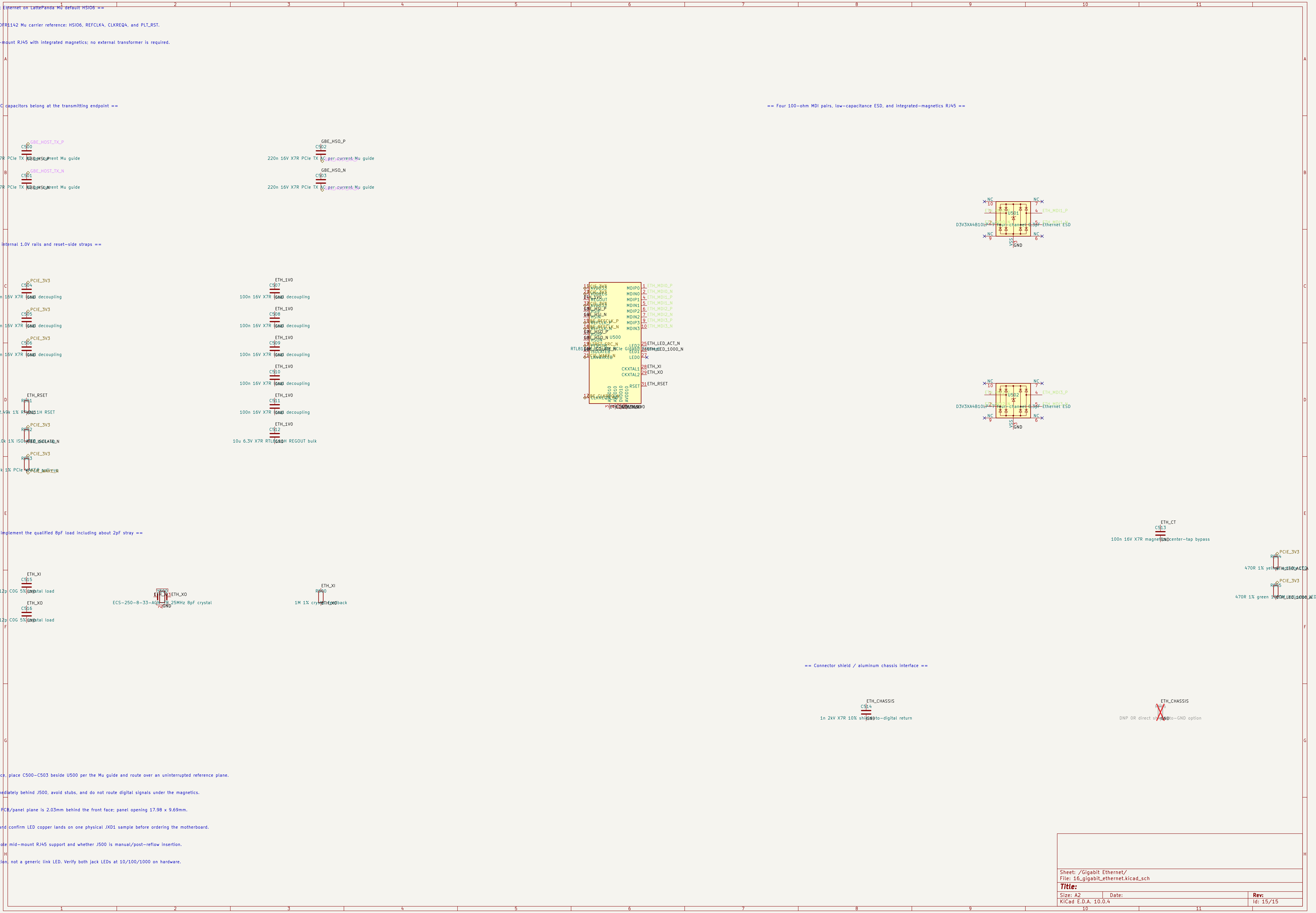
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 13/15


rt USB hub, playback/record codec, microphone, and stereo BTL amplifier ==
multi-TT hub. Port 1 is the existing radio codec; port 2 is the PCM2900C system codec.
le. TPS2052B implements host-controlled VBUS and individual overcurrent feedback.
ted 5V audio branch and downstream USB power ==
-TR: self-powered, strap-configured, both ports non-removable ==
stem USB playback/record codec; TI Figure 38 bus-powered core ==
== Pop-safe amplifier enable: PCM2900 operational AND EC permission ==
RTJR, 12dB gain, TI single-ended input and EMI networks ==
100n AUDIO_5V high-frequency bypass
TPS2052BD
1u TPS2052B input
10u radio-codec VBUS
100u 10V polymer downstream VBUS bulk
ABM8-24.000MHZ-10-1-U-T 24MHz 10pF CL
12.0k 1% USB transceiver RBIAS
TLV803EA29RDBZR 2.93V reset supervisor
10k NON_REM1 strap high
100k NON_REM0 strap low
100k CFG_SEL1 strap low
ABM8G-106-12.000MHZ-T 12MHz 10pF CL
18p C0G 5% codec crystal load
1.5k USB D+ pull-up per TI Figure 38
PCM2900 USB Audio Class stereo codec
SN74LVC1G08 PCM2900 SSPND AND EC audio enable
100n amplifier supply at AVDD/PVDD
TPA2012D2 stereo BTL class-D amplifier
MPZ1608S221A 220R@100MHz speaker EMI bead
Sheet: /System Audio/
File: 15_system_audio.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 14/15



e Gigabit Ethernet on LattePanda Mu default HSIO6 ==
Panda's DFR1142 Mu carrier reference: HSIO6, REFCLK4, CLKREQ4, and PLT_RST.
ole mid-mount RJ45 with integrated magnetics; no external transformer is required.
erface; AC capacitors belong at the transmitting endpoint ==
== Four 100-ohm MDI pairs, low-capacitance ESD, and integrated-magnetics RJ45 ==
220n 16V X7R PCIe TX AC per current Mu guide
H 3.3V / internal 1.0V rails and reset-side straps ==
100n 16V X7R decoupling
10u 6.3V X7R RTL8111H REGOUT bulk
D3V3XA4B10LP 4-channel 0.35pF Ethernet ESD
U500
U501
U502
2pF legs implement the qualified 8pF load including about 2pF stray ==
12p C0G 5% crystal load
ECS-250-8-33-AGN 25MHz 8pF crystal
1M 1% crystal feedback
100n 16V X7R magnetics center-tap bypass
470R 1% yellow ETH_LED_ACT_N
470R 1% green ETH_LED_1000_N
== Connector shield / aluminum chassis interface ==
1n 2kV X7R 10% shield-to-digital return
DNP 0R direct shield-to-GND option
PCIe device, place C500–C503 beside U500 per the Mu guide and route over an uninterrupted reference plane.
U502 immediately behind J500, avoid stubs, and do not route digital signals under the magnetics.
1.30mm; PCB/panel plane is 2.03mm behind the front face; panel opening 17.98 x 9.69mm.
ommon and confirm LED copper lands on one physical JXD1 sample before ordering the motherboard.
hrough-hole mid-mount RJ45 support and whether J500 is manual/post-reflow insertion.
M indication, not a generic link LED. Verify both jack LEDs at 10/100/1000 on hardware.
Sheet: /Gigabit Ethernet/
File: 16_gigabit_ethernet.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 15/15